Yash Mathur

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PROFILE

PhD researcher building fast, deployable ML for drug discovery. My focus is on closing the gap between research-grade models and tools that scientists actually use in practice. Co-developed InstaDock, a virtual screening platform now with 300+ citations and adopted by labs worldwide. Built VeloBind, a binding affinity predictor that uses only protein sequences and molecular fingerprints, no 3D structure required yet outperforms every sequence-based DTA predictor on the CASF-2016 and CASF-2013 benchmarks. UGC NET qualified in Computer Science (2024). Proficient in Python and PyTorch, with hands-on experience in REST API design, GUI development with PyQt and Flask, and containerized deployment via Docker.

SKILLS

- **Languages**: Python (Expert), R
- **Deep Learning & AI**: PyTorch, TensorFlow, Scikit-learn, GANs, Protein Language Models
- **Data Science**: Pandas, NumPy, SciPy, Matplotlib, Seaborn, Plotly
- **Dev & DevOps**: Git, Docker, Linux/Bash, REST API design
- **GUI & Web**: Flask, PyQt5
- **Domain-Specific**: Biopython, RDKit, OpenBabel, Chemprop, MDAnalysis

EXPERIENCE

Project Assistant *Dr. Hassan's Lab, Jamia Millia Islamia | 2020 – 2023*

- Developed **InstaDock** and **PyPAn**, two automated Python-based tools that accelerated bioinformatics workflows by up to 100x.
- Engineered a GUI-based batch docking pipeline using PyQt and QuickVina-W, reducing typical 15-25 minute manual workflows to 1-2 minutes.
- Consolidated complex protein analysis workflows involving energy minimization and structural assessment into a unified software suite, eliminating the need for multiple separate programs.

PROJECTS

1. VeloBind: A structure-free protein-ligand binding affinity predictor intended for primary screening
(Status: Under review)

- Built a binding affinity predictor using ESM2-35M embeddings and molecular fingerprints, no 3D structure required, enabling inference on thousands of compounds in seconds on a single GPU.
- Benchmarked on CASF-2016 and CASF-2013. Outperforms every sequence-based DTA predictor. Ranks second overall, behind only HPDAF, which relies on full 3D pocket geometry.

2. Generative AI for De Novo Ligand Design
(Status: Ph.D. Thesis)

- Developing a large-scale generative framework leveraging Transformer and VAE architectures to explore chemical latent space beyond existing libraries.
- Implemented multi-objective optimization to condition molecule generation on ADMET constraints, ensuring high drug-likeness and synthesizability.

3. OmniDock: Next-Generation Automated Screening Platform
(Status: Work-in-progress)

- Engineered a unified, end-to-end virtual screening platform integrating multiple docking engines under a single interface, covering blind docking, site-specific multi-scoring, covalent, flexible sidechain, macrocycle, and protein-protein docking workflows without requiring users to switch tools.
- Automated the entire pipeline from ligand procurement via ZINC, PubChem, and IMPPAT APIs through file preparation, large-scale batch docking, hit identification, and result generation, producing structured CSV outputs with binding affinity, pKi, ligand efficiency, and QED alongside automated PDF reports with 3D visualizations.

EDUCATION

PhD in Computational Biology (Expected December 2026)
Jamia Millia Islamia | New Delhi, India
Thesis title: "Employing Generative Deep Learning Models for Strategic Advancements in Drug Development"

Master of Science in Bioinformatics (2020)
Jamia Millia Islamia | New Delhi, India
Dissertation title: "Modelling Method Advisory Tool: a web-based module for protein modelling advise"

Bachelor of Science (H) in Microbiology (2018)
University of Delhi | New Delhi, India

SELECTED PUBLICATIONS

- Mathur, Y., et al. "VeloBind: A structure-free protein-ligand binding affinity predictor intended for primary screening" Submitted for Publication (Manuscript Under Review).
- Mathur, Y., et al. (2025). "Current advancement in AI-integrated drug discovery: Methods and applications."; Biotechnology Advances.
- Mohammad, T., Mathur, Y., et al. (2021). "InstaDock: A single-click graphical user interface for molecular docking-based virtual high-throughput screening"; Briefings in Bioinformatics. (300+ citations)